Engineering Structures 274 (2023) 115156

Contents lists available at ScienceDirect

Engineering Structures

journal homepage: www.elsevier.com/locate/engstruct

Prediction of FRP-concrete interfacial bond strength based on machine learning

Feng Zhang [a], Chenxin Wang [a], Jun Liu [b], Xingxing Zou [a,*], Lesley H. Sneed [c], Yi Bao [d], Libin Wang [a]

[a] College of Civil Engineering, Nanjing Forestry University, Nanjing 210037, China
[b] Louisiana Transportation Research Center, Louisiana State University, Baton Rouge, LA, USA
[c] Department of Civil, Materials and Environmental Engineering, University of Illinois at Chicago, 929 W. Taylor St, Chicago, IL 60607, USA
[d] Department of Civil, Environmental and Ocean Engineering, Stevens Institute of Technology, Hoboken, NJ 07030, USA

ARTICLE INFO

Keywords:
Bond strength
extreme gradient boosting (XGBoost)
fiber reinforced polymer (FRP)
Isolation forest
Interpretable machine learning (ML)
Random forest (RF)

ABSTRACT

Externally bonding fiber reinforced polymer (FRP) to concrete structures is an effective way to enhance the mechanical performance of concrete structures. Many equations have been proposed to predict the interfacial bond strength for FRP-concrete structures but have limited accuracy due to the complexity of the bond behavior. This study proposes to formulate the FRP-concrete interfacial bond strength based on machine learning (ML) methods, which have emerged as a promising alternative to achieve high prediction accuracy in high-dimension problems. To this end, a database containing 1,375 FRP-concrete direct shear test specimens that failed due to interfacial debonding was established. The database was improved using an unsupervised isolation forest that identified and eliminated anomalous data, and was then used to train six ML models, namely artificial neural networks (ANN), support vector machine, decision tree, gradient boosting decision tree, random forest, and XGboost algorithms, to predict the FRP-concrete interfacial bond strength. The ML predictive models showed higher accuracy than 16 existing equations in the literature. The XGBoost model showed the highest accuracy, and its coefficient of variation was 54% lower than the existing equation with the highest accuracy among those considered. The ANN model was used to perform a parametric study on the influencing parameters, and a new equation was generated to predict the interfacial bond strength, considering the key influencing parameters. The equation enables interpretation of the ML models. The study combines ML models and traditional physical models to achieve a novel, interpretable ML method for predicting FRP-concrete interfacial bond strength.

1. Introduction

Reinforcement corrosion and concrete degradation are typical deficiencies of concrete structures that compromise the strength and durability [1–5]. Fiber reinforced polymer (FRP) is a promising material for repairing concrete structures, and FRP features light weight, high strength, and high corrosion resistance. Externally bonding FRP to concrete structrures is an effective way to increase the flexural, shear, axial, and seismic resistance of structures [6–11]. However, field applications [7] showed that FRP-strengthened concrete was prone to interfacial debonding that was able to trigger structural failure at low load levels.

Intensive research has been conducted on the prediction of the FRP-concrete interfacial strength [12–15]. Since the 1990 s, a number of equations have been proposed to quantify the FRP-concrete interfacial bond strength. Before 2000, most researchers calculated the interfacial bond strength based on an average bond strength multiplied by the area of the interface, assuming that the shear stress is distributed evenly across the bonded length and width directions [16–19]. After 2000, the stress transition zone was researched, and the concept of the effective bond length, i.e., the minimum length required to establish the load-carrying capacity of the interface due to bond, was introduced and improved the prediction accuracy [18–23]. Then, by considering the nonuniform stress distribution along the width direction, several scholars further improved the prediction accuracy by introducing a width factor [1,22,24–28]. Recently, Zhang et al. [29] analyzed the mixed-mode debonding of the FRP–concrete interface by a cohesive zone model (CZM), derived from the modified Mohr–Coulomb strength criterion, and provided a detailed description of the initial debonding

* Corresponding author.
E-mail address: civilzou@njfu.edu.cn (X. Zou).

https://doi.org/10.1016/j.engstruct.2022.115156

Available online 4 November 2022

Nomenclature

ANN	artificial neural network;
Avg	average value;
b_f	width of the fiber strip;
b_c	width of the concrete blocks/prism;
CoV	coefficient of variation;
DT	decision tree;
E_f	elastic modulus of the fiber strips;
FRP	fiber reinforced polymer;
F_1	function of K_f in Eq. (10);
F_2	function of $f_c^{'}$ in Eq. (10);
F_3	function of L in Eq. (10);
F_4	function of $\frac{b_f}{b_c}$ in Eq. (10);
$f_c^{'}$	compressive strength of concrete;
f_t	tensile strength of concrete;
G_1	fitting equation for K_f and $\frac{P_u}{b_f}$;
G_2	fitting equation for f_c' and $\frac{P_u}{b_f}$;
G_3	fitting equation for L and $\frac{P_u}{b_f}$;
G_4	fitting equation for $\frac{b_f}{b_c}$ and $\frac{P_u}{b_f}$;
G_F	fracture energy of interface;
GBDT	gradient boosting decision tree;
IAE	integral absolute error;
K	number of base learners;
K_f	stiffness of the fiber strip;
L	bonded length;
L_e	effective bond length;
L_ϵ	ϵ-insensitive loss function;
ML	machine learning;
P_u	interfacial bond strength;
$P_{u\_Model\_i}$	predicted value of the interfacial bond strength;
$P_{u\_t\_i}$	test result of the interfacial bond strength;
P_k, $P_{k^{'}}$	probability that the sample belongs to class k or k';
RF	random forest;
RMSE	root mean square error;
SVM	support vector machine;
T	sample set;
t_f	thickness of the fiber strip;
XGBoost	extreme gradient boosting;
w_{ki}	weight of neurons;
x_i	input signal;
y	number of decision trees;
y_k	output signal;
α_i, μ_i	Lagrange multipliers;
α_1, α_2, α_3, α_4, β_2, β_3, β_4	parameters of Eq. (10);
β_L	parameter of bond length;
ϵ	acceptance deviateon;
θ	threshold of the neuron;
κ_w	width factor;
τ_α	interfacial shear stress;
Ω	function used for regularization to avoid overfitting.

and propagation of the FRP–concrete interface through a full range mixed-mode debonding process. Zhang and Huang [30] analyzed the mixed-mode debonding behavior of the FRP-concrete interface by a 3D-CZM. The interfacial damage propagation and friction-sliding evolution during the whole debonding process have been gradually clarified, and the prediction accuracy of related equations has been improved compared with earlier equations. However, the developed equations and methods still have limited accuracy because they did not account for (i) the unavoidable errors of test data or simulation results caused by the scatter of concrete strength and the multi-scale characteristics of concrete, and (ii) the multi-variable correlation between the many influencing parameters such as the geometric size and material properties of the test specimens [31]. Prediction of the bond strength is a high-dimension problem involving many variables.

To address the above challenges and account for the variation of experimental data, machine learning (ML) has emerged as a promising alternative with high prediction accuracy. The representative ML algorithms include artificial neural network (ANN), support vector machine (SVM), random forest (RF), and extreme gradient boosting (XGBoost). Su et al. [32] employed three different ML methods to predict interfacial bond strength considering the correlation between influencing variables and interfacial bond strength. Jahangir and Rezazadeh [33] proposed an equation for interfacial bond strength based on ANN with the artificial bee colony optimization. Vu and Hoang [34] predicted the punching shear capacity of FRP-reinforced slabs based on SVM and the firefly optimization. Those studies considered only existing data in training the ML models, without considering the amechanical behaviors. Recently, Mahjoubi et al. [35] proposed a logic-guided neural network, which integrated data-driven and mechanical approaches to predict the interfacial bond strength between steel and concrete as well as the bond-slip curves. A novel loss function that considered logic based on mechanics was presented, and logic-guided datasets were created. The results showed that the proposed approach achieved high prediction accuracy. In addition to research on interfacial bond, studies were conducted to predict and design advanced fiber-reinforced concrete such as ultra-high-performance concrete [36] and strain-hardening cementitious composites ([37,38]).

Much of the existing research on ML models showed variable prediction accuracy against different test databases with less than 1000 specimens. A large database of test results is essential to reveal the variation of the database accounting for scattering of material properties and test methods. In addition, most ML approaches have been based on "black-box" models, which do not support decision-making in a way that humans are able to explicitly explain. There are limitations in applying the "black-box" models to engineering practices. Currently there is limited research on interpretable ML approaches to predict the FRP-concrete interfacial bond strength. Thus, there is an urgent need to provide a method to fill the gap in the practicability and interpretability of ML models for this purpose.

This paper presents an interpretable ML approach to predict the FRP-concrete interfacial bond strength and to support the design of FRP-strengthened concrete structures. This research has three main technical contributions: (1) A large database, consisted of test results from 1,375 FRP-concrete direct shear specimens, was established by integrating experimental test data from different studies. It is the first time that steel cords reinforced polymers (SRP)-, and scrimber-concrete specimens are included. (2) The performance of six ML algorithms, namely ANN, SVM, decision tree (DT), gradient boosting decision tree (GBDT), RF, and XGboost, was evaluated and compared. (3) A new equation, with interpretable physical meanings for the parameters, was formulated to predict the interfacial bond strength by considering the influencing parameters identified from the ML models. Results were compared with those from 16 existing equations in the literature. This research advances the capabilities of predicting the FRP-concrete interfacial bond properties, revealing interpretability of the proposed ML models, and deriving a practical prediction equation that can combine ML models and traditional physical meaning-driven analysis methods.

2. Database of FRP-concrete direct shear test specimens

A variety of methods have been proposed to test the FRP-concrete interfacial shear performance. The direct shear test is a popular test method, as shown in Fig. 1. The direct shear test includes single-lap or double-lap shear tests. The material properties, geometric parameters, and test values of interfacial bond strength, P_u, are typically reported for each test. Based on previous experiments, the results of P_u are influenced by eight predominant parameters, which are the bonded length of the fiber strip L, the compressive and tensile strengths of concrete $f_c^{'}$ and f_t, the widths of the concrete block and fiber strip b_c and b_f, the modulus of fiber strip E_f, and the thickness of the fiber strip t_f. It should be noted $E_f t_f$ is a measure of the stiffness of the FRP strip and is always used together in the prediction models reported in the literature. $E_f t_f$ is denoted by K_f hereafter. For each specimen, the influence parameters are (L, $f_c^{'}$, f_t, b_c, b_f, K_f), and the result is P_u.

This research assembled a database of data collected from 1,375 specimens tested using single- or double-lap shear tests from 51 references ([18,26,28,39–86]), presented in the supplementary materials of this paper, according to the following criteria:

(1) The specimen was tested under monotonic loading. Specimens tested under cyclic or fatigue loading were excluded.
(2) The FRP strip materials included fibers made of carbon (CFRP), glass (GFRP), aramid (AFRP), steel cords (SRP), or scrimber bonded to concrete by polymeric adhesive. It should be noted that scrimber is a eco-friendly material consisting of bamboo fibers and epoxy, normally with E_f of 10–15 GPa. More details are available in Refs. [75,76]. Specimens with steel plates or other fiber types were excluded.
(3) Specimens with external anchorage systems were excluded.
(4) The specimen failed due to interfacial debonding. Specimens exhibiting the fiber rupture failure mode were excluded.

Different loading rates and other controlling factors of the loading were not differentiated, although they might influence the results [87]. This database is the first database that incorporates the test data of SRP (with $E_f \geq 250$ GPa) and scrimber (with $10 \text{ GPa} \leq E_f \leq 15$ GPa) bonded to concrete.

Around the world, concrete material strength is tested using different control specimens. In order to compare the results from different studies, this research converted the concrete compressive strength values determined using different control specimens to the compressive strength of a cylinder $f_c^{'}$ [88]. Compressive strength values determined by concrete cubes and prisms, f_{cu}, were converted to $f_c^{'}$ according to the relationship in Eq. (1a) [89]. Differences in cube and prism sizes [90,91] were not differentiated since they are generally very close – usually within 5% [13]. Regarding the concrete tensile strength, f_t, reported values were used directly. For studies in which concrete tensile strength was not reported, Eq. (1b), proposed in [92] and used in [72], was adopted herein to calculate f_t from $f_c^{'}$. The difference in tensile tests types was not differentiated.

$$f_c^{'} = 0.78 f_{cu} \tag{1a}$$

$$f_t = \begin{cases} 0.3 \times \sqrt[3]{(f_c{'})^2} \ , f_c^{'} \leq 60 \text{ MPa} \\ 2.12 \times ln\left(1 + \frac{f_c^{'}}{10}\right) \ , f_c^{'} > 60 \text{ MPa} \end{cases} \tag{1b}$$

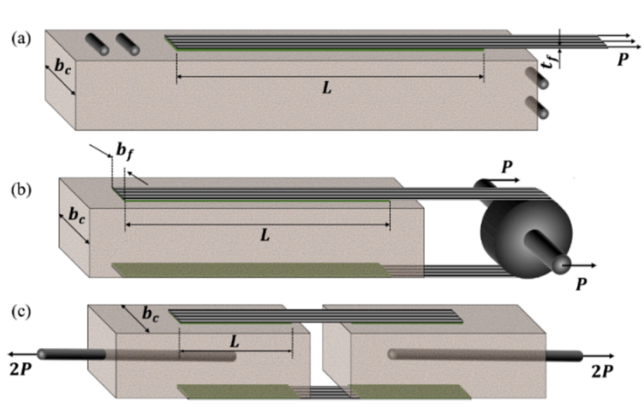

Fig. 1. Typical direct shear setup for an FRP-concrete joint: (a) single-lap, (b) double-lap with one prism, and (c) double-lap with double prisms.

3. Methods

3.1. Anomaly detection

The raw database was evaluated to identify and eliminate anomalous data instances [93]. This study employed an unsupervised algorithm - isolation forest [94] to detect anomalous data, illustrated in Fig. 2**.** The algorithm evaluates data by an ensemble of decision trees and gives an anomaly score in the interval 0–1. When the anomaly score is greater than a certain threshold, the data is considered as anomalous data. The ratio of the anomalous data to the total dataset is defined as the contamination ratio ([36–38]). The black dots in Fig. 2 represent the root nodes of isolation trees. The dashed circles are the investigated data point. The other dots are data points. The gray lines represent the path length. The isolation forest algorithm was implemented in Statistics and Machine Learning Toolbox of Matlab.

Fig. 3 shows the effect of contamination ratio on the prediction error of the XGBoost for the data in the database. Fig. 3 shows the effect of contamination ratio on the prediction error of the XGBoost for the data in the database. XGBoost was selected for this purpose based on results of a preliminary investigation that showed it performed better than the other algorithms considered. The root mean square error (RMSE) was normalized to a range of 0–1 and designated as normalized RMSE (NRMSE). Deleting anomalous data significantly reduced the NRSME, which was minimized when the contamination ratio was 25%, meaning that 25% data were deleted (i.e., 345 data), and 1030 data were kept. The threshold defined as anomalous data by isolation forest was 0.4886.

3.2. Input distribution and analysis

The distribution ranges of the six influencing parameters (L, $f_c{'}$, f_t, b_c, b_f, K_f) and the result, P_u, were analyzed, as shown in Fig. 4. The raw database shown in the supplementary materials of this paper included 1375 specimens, and the range of each parameter is shown in Fig. 4**a**. The scale of the raw database was reduced to 1030 specimens after deleting specimens according to anomaly detection method as discussed in **Section 3.1**. Table 1 shows the ranges of each parameter after the data were reduced, and this reduced database will be used in the remainder of this paper. The comparison between Fig. 4**a and 4b** indicates that many specimens with higher P_u were deleted according to anomaly detection method.

Fig. 5 shows the results of the Pearson correlation coefficient analysis of the six parameters (L, $f_c^{'}$, f_t, b_c, b_f, K_f). The degree of correlation between f_t and $f_c^{'}$ reaches 0.931, which is expected since Eq. (1b) was used to determine the value of f_t for most (57.8%) specimens. Thus, it can be concluded that $f_c^{'}$ is able to represent the concrete properties, and f_t was not considered in the training of ML models to reduce dimensionality of the influencing parameters and avoid the multicollinearity issues ([36–38]).

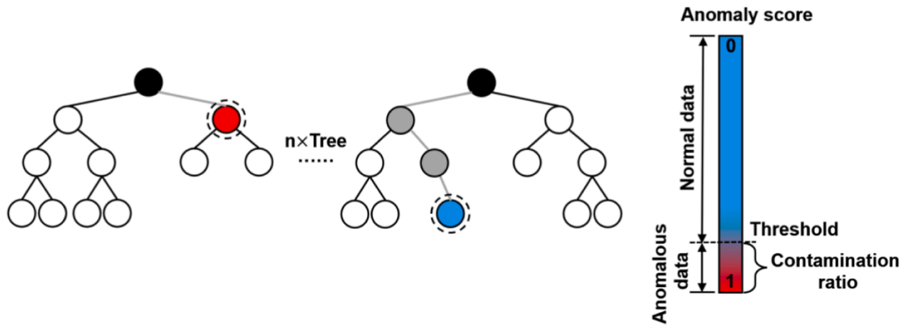

Fig. 2. Illustration of anomaly detection with isolation forest.

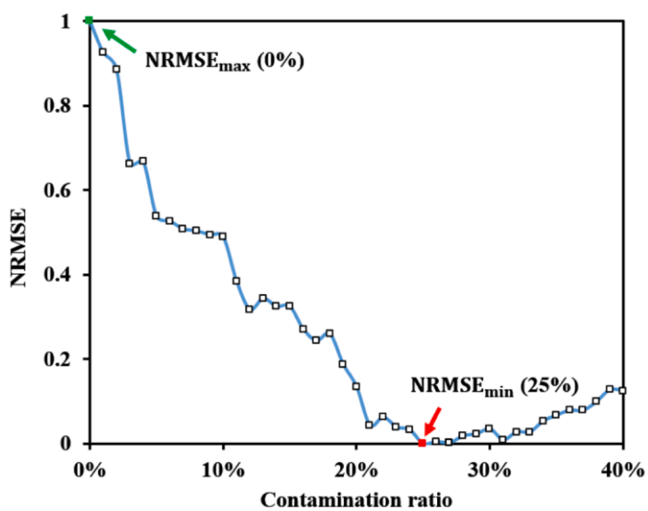

Fig. 3. Effect of contamination ratio on the prediction errors of XGBoost.

3.3. Machine learning algorithms

This study investigated the performance of six representative ML algorithms, which are ANN, SVM, DT, GBDT, RF, and XGBoost, for prediction of the FRP-concrete interfacial bond strength. The six algorithms were elaborated in Refs. [36–38]. Description of these algorithms are not duplicated in this paper.

3.3.1. ANN

ANN imitates the generation mechanism of resting and action potentials of human brain neurons. Fig. 6**a** shows the architecture of an ANN. The output is given by Eq. (2):

$$y_k = F\left(\sum_{i=1}^{n} w_{ki}x_i - \theta\right) \tag{2}$$

where y_k is the output signal, w_{ki} is the connection weight of the neuron, x_i is the input signal, and θ is the threshold of the neuron, which is commonly known as the bias term. The neurons of adjacent layers are

Table 1
The ranges of input and output variables in the reduced database.

Parameters	Minimum value	Maximum value
P_u (kN)	3.8	40.3
L (mm)	20	400
f_c' (MPa)	15.2	74.7
f_t (MPa)	1.8	4.6
b_c (mm)	80	200
b_f (mm)	15	100
K_f (GPa · mm)	11.9	238.0

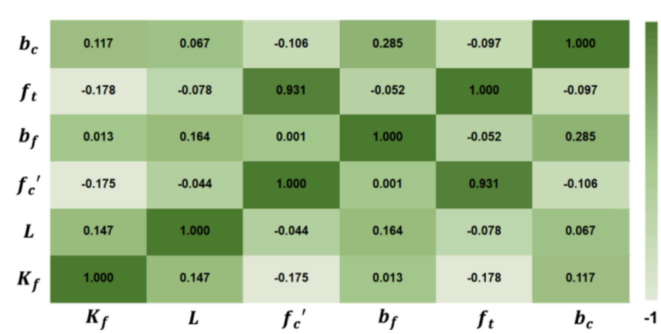

Fig. 5. Pearson correlation coefficient.

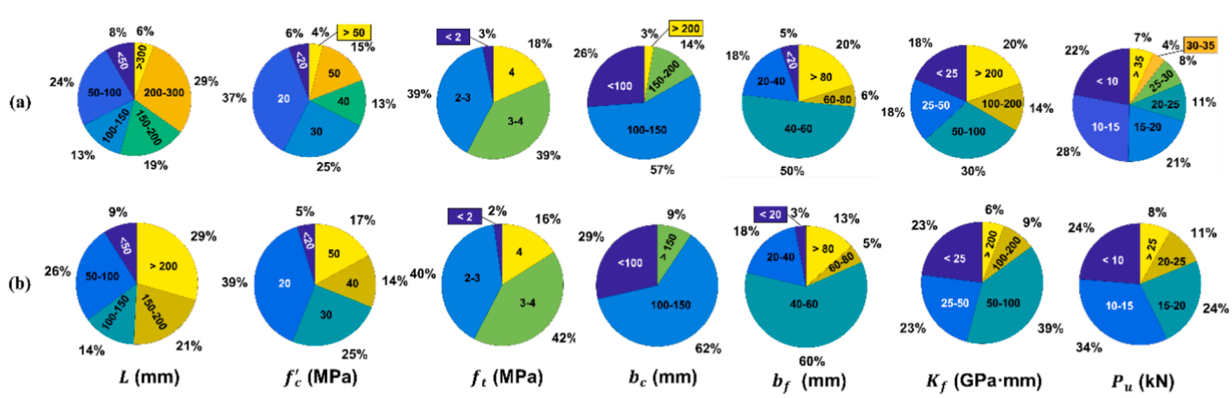

Fig. 4. The distribution of input and output variables: (a) the raw database with 1375 specimens, and (b) the reduced database with 1030 specimens.

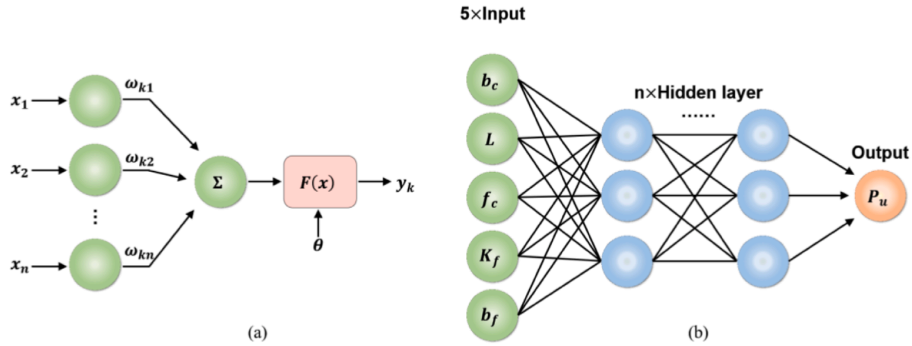

Fig. 6. ANN: (a) artificial neuron, and (b) multi-layer feedforward neural networks.

interconnected, as shown in Fig. 6**b**. In this study, five input parameters (L, f_c', b_c, b_f, K_f) and one output parameter, P_u, were used. In order to determine the number of hidden layers and hidden neurons, the trial-and-error method was used. The number of hidden layers and hidden neurons were set to 1 and 80, respectively. The transfer functions of the hidden layers and the output layer were "Tansig" and "Purelin", respectively. The Levenberg-Marquardt algorithm with high convergence efficiency was selected as the training function [95].

3.3.2. SVM

SVM is able to solve regression problems whose model output $f(x)$ is not exactly the same as the real output y. As shown in Fig. 7**a**, given the training sample $X = \{(x_1, y_1), (x_2, y_2), \cdots\cdots(x_n, y_n)\}$, aiming to obtain $f(x) = \omega^T x + b$, the prediction is correct if y is within $f(x) \pm \epsilon$, and the loss is calculated only when the error between $f(x)$ and y exceeds ϵ. Symbols ω and b are the model parameters to be determined. SVM determines the regression function by minimizing the objective function, introducing slack variables ξ_i and $\widehat{\xi}_i$, which depend on L_ϵ, i.e., ϵ-insensitive loss function. The expression of L_ϵ is shown as Eq. (3) and Fig. 7**b**. In this study, the Radial Basis Function (RBF) was used as the kernel function because RBF realizes nonlinear mapping and has fewer parameters. The number of parameters affects the complexity of the model [96].

$$L_\epsilon(z) = \begin{cases} 0, & if\ |z| \le \epsilon \\ |z| - \epsilon, & otherwise \end{cases} \tag{3}$$

3.3.3. DT

DT intuitively uses probability analysis by forming a decision tree to obtain the probability that the expected value of the net present value is greater than or equal to zero. DT is capable to evaluate the project risk and judge the feasibility based on the known probability of occurrence of various situations [97]. DT has been used in statistics and data mining. The cart decision tree uses Gini index, $Gini(T)$, to select partition attributes, and the purity of database t is measured by the Gini index:

$$Gini(T) = \sum_{k=1}^{|y|} \sum_{k' \ne k} P_k P_{k'} = 1 - \sum_{k=1}^{|y|} P_k^2 \tag{4}$$

where T is sample set, P_k and $P_{k'}$ are the probability that the sample belongs to class k or k', respectively, and y is the number of decision trees.

3.3.4. GBDT

GBDT consists of multiple decision trees, and the conclusions of all trees are added to determine the final answer. The core idea is to use negative gradient approximation to simulate the residual. GBDT is conducted using the following steps:

Step 1: Initialize the weak learner;

Step 2: Calculate the value of the negative gradient of the loss function in the current model, using it as an estimate of the residuals;

Step 3: Estimate the regression leaf node area to fit an approximation of the residuals;

Step 4: Use linear search to estimate the value of the leaf node region to minimize the loss function;

Step 5: Update the regression tree.

3.3.5. RF

RF is a classifier that contains multiple decision trees, and the category of its output is determined by the mode of the category output by the individual tree. When a sample needs to be predicted, the predictions of each tree in the forest for that sample are counted, and then the final result is selected from these predictions by voting, as shown in Fig. 8.

3.3.6. XGBoost

XGBoost is an algorithm or engineering implementation based on GBDT [98]. The basic idea of XGBoost is the same as GBDT, but XGBoost makes the following optimizations:

(1) Perform a second-order Taylor expansion of the loss function to improve the accuracy.

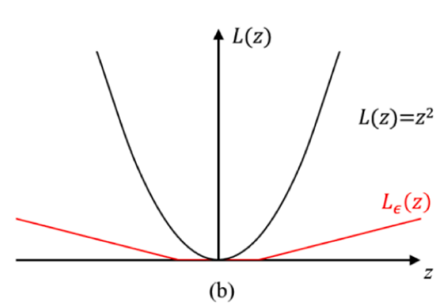

Fig. 7. SVM: (a) regression diagram, and (b) ϵ-insensitive loss function.

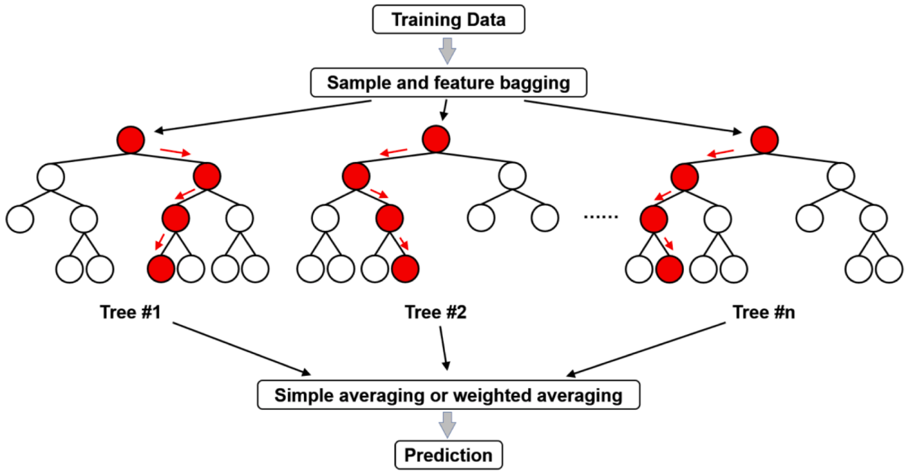

Fig. 8. Architecture of RF regression prediction model.

(2) Use regular terms to control the complexity of the model and avoid overfitting.
(3) Use Blocks storage structure for parallel computing.

3.4. Performance metrics

Four performance metrics were used to evaluate the ML models and the equations, which are RMSE, coefficient of variation (CoV), average value (Avg), and integral absolute error (IAE): RMSE is used to indicate how much error the model will produce in predictions, with higher weights for larger errors; CoV is an indicator of the relative scatter of the values; Avg indicates the average performance of the overall forecast compared to the original data; IAE represents the prediction accuracy of the model from the perspective of relative error, avoiding the influence of the size of the value itself. The four indicators are classic indicators for evaluating the prediction accuracy, which can intuitively reflect the error, discreteness, and accuracy of the prediction results, and can comprehensively evaluate each model.

$$\mathrm{RMSE} = \sqrt{\frac{\sum_{i=1}^{n}\left(P_{u\_Model\_i} - P_{u\_test\_i}\right)^2}{n}} \tag{5}$$

$$\mathrm{CoV} = \frac{\sqrt{\frac{\sum_{i=1}^{n}\left(\frac{P_{u\_Model\_i}}{P_{u\_test\_i}} - \mathrm{Avg}\right)^2}{n}}}{\mathrm{Avg}} \tag{6}$$

$$\mathrm{Avg} = \frac{\sum_{i=1}^{n}\frac{P_{u\_Model\_i}}{P_{u\_test\_i}}}{n} \tag{7}$$

$$\mathrm{IAE} = \sum_{i=1}^{n}\frac{\left|P_{u\_Model\_i} - P_{u\_test\_i}\right|}{\sum P_{u\_test\_i}} \tag{8}$$

where $P_{u\_Model\_i}$ and $P_{u\_test\_i}$ are the predicted and experimental values of the interfacial bond strength, P_u, respectively; and n is the number of specimens considered.

4. Evaluation of existing prediction equations

4.1. Prediction equations

Researchers have proposed many prediction equations for interfacial bond strength based on tests ([1,24]) or numerical simulation [19]. As discussed in **Section 1**, equations based on the average shear stress had lower prediction accuracy until the concept of effective bond length was proposed. After introducing the width effect and fracture energy, the accuracy of the equations was further improved. This study evaluated 16 existing equations, listed in **Appendix**. The prediction equations are written as:

$$P_u = b_f \kappa_w \beta_L L Y\left(G_F, K_f\right) \tag{9}$$

where κ_w is a width factor related to the width of FRP strip b_f and the width of concrete block b_c; β_L is a parameter that reflects the influence of effective bond length L_e and L; and $Y(G_F, K_f)$ is a term related to K_f and the fracture energy G_F.

4.2. Comparison of prediction equations

Fig. 9 shows the prediction results of the specimens in the reduced database by the 16 existing equations in **Appendix**, along with the same metrics used to represent the accuracy of the results of the ML models. The green solid line in the figures indicates that the prediction results is exactly the same as the actual value. The red dotted line indicates the overall prediction trend of the equations. The closer the red line is to the green line, the higher the prediction accuracy. The black dots represent the prediction of each sample value. When they are all distributed around the green line, it means the prediction error is small. The results show that of the equations proposed, the prediction results of Model_1 by Holzenkämpfer [99], Model_2 by Tanaka [16], and Model_7 by Niedermeier [100] are all conservative (i.e., the predicted values are generally less than the test values), which may be due to the fact that these equations were proposed earlier, when most researchers calculated the interfacial bond strength based on the average bond strength multiplied by the interfacial area, and the materials used in the tests at that time could not reach the current high strength. As discussed in **Section 1**, with the introduction of effective bond length and width effects, the accuracy of the prediction equation is significantly improved. For instance, Model_8 by Yang et al. [21], Model_9 by Chen and Teng et al. [1], Model_14 by Lu et al. [24], and Model_15 by Wu et al. [25] have higher accuracy (Avg ≈ 1.0) and lower variation (smaller RSME, CoV, and IAE).

5. Performance evaluation

5.1. Prediction performance

With the database modified using the anomaly detection method, 85% of the reduced database (875 data instances) were randomly selected as the training set, and 15% of the reduced database (155 data instances) were used as the test set. Table 2 shows the performance

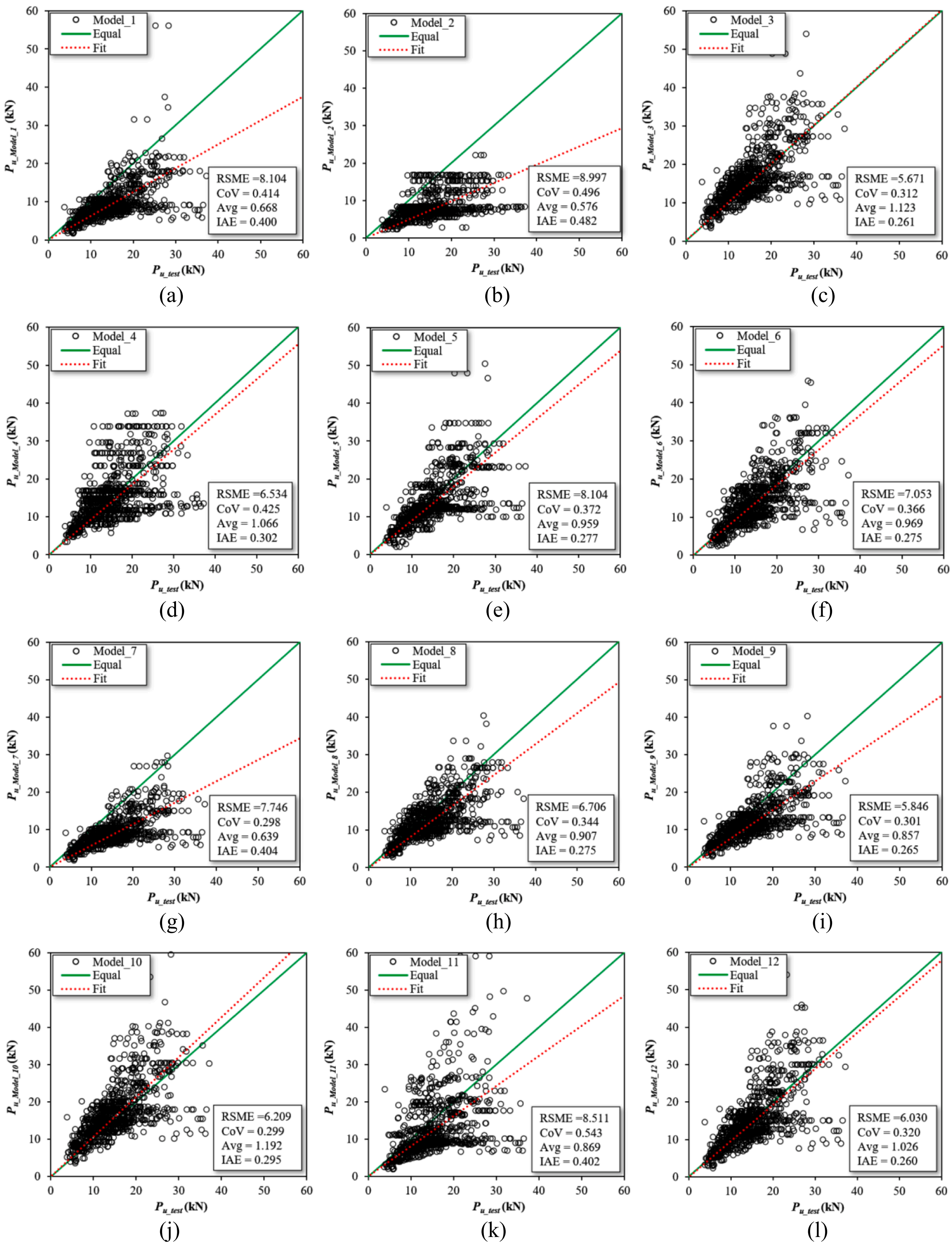

Fig. 9. Prediction performance of 16 existing equations: (a) Model_1, (b) Model_2, (c) Model_3, (d) Model_4, (e) Model_5, (f) Model_6, (g) Model_7, (h) Model_8, (i) Model_9, (j) Model_10, (k) Model_11, (l) Model_12, (m) Model_13, (n) Model_14, (o) Model_15, and (p) Model_16.

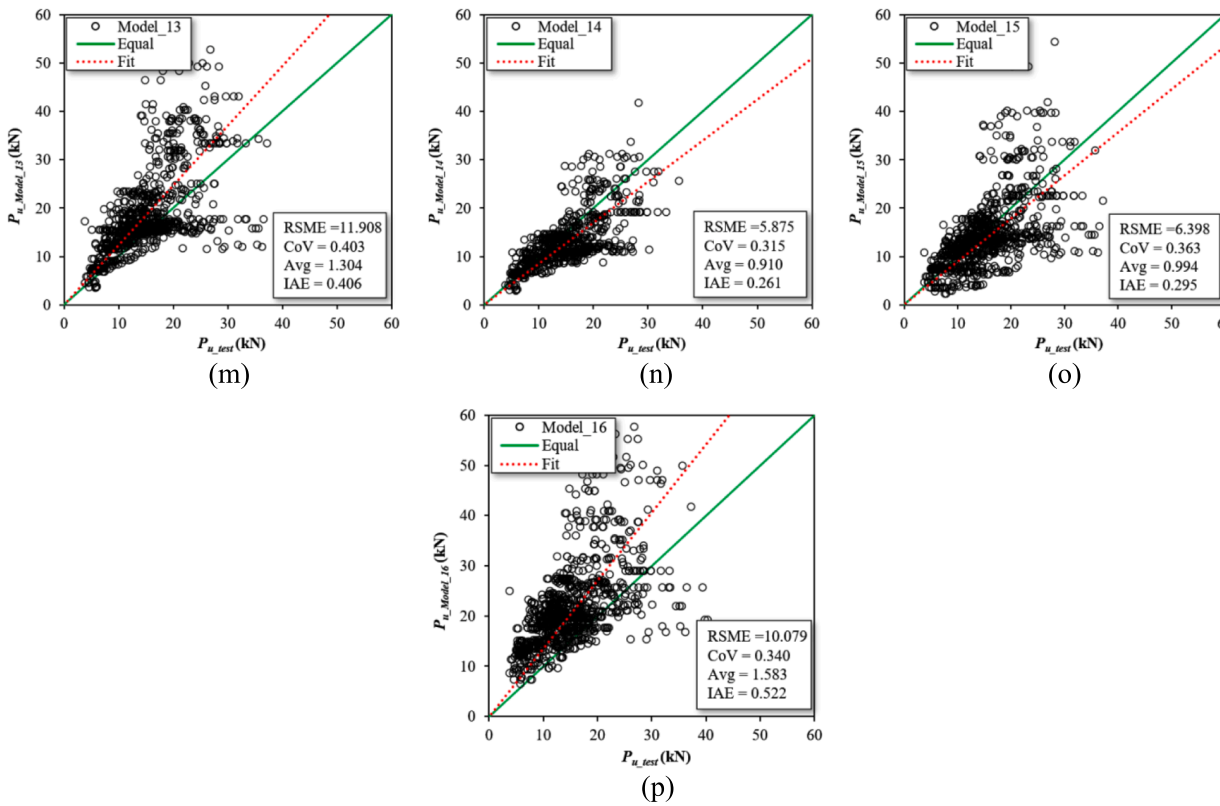

Fig. 9. (*continued*).

Table 2
Performance metrics of the trained ML models.

ML model	Set	RSME	CoV	Avg	IAE
ANN	Training	1.653	0.125	1.013	0.072
	Test	3.668	0.224	1.062	0.157
SVM	Training	3.349	0.210	1.029	0.116
	Test	2.882	0.201	1.053	0.024
DT	Training	2.516	0.186	1.030	0.123
	Test	3.062	0.224	1.046	0.146
GBDT	Training	1.402	0.107	1.011	0.052
	Test	2.426	0.151	1.030	0.111
RF	Training	2.247	0.170	1.033	0.109
	Test	2.805	0.190	1.056	0.133
XGBoost	Training	1.377	0.104	1.009	0.047
	Test	2.528	0.157	1.030	0.112

metrics of the trained ML models.

In order to compare ML models with the 16 existing equations (discussed further in **Section 5.2**), the entire reduced database (1,030 data) was used instead of being divided into training and test sets. The comparison of the predicted data of the trained ML models and the entire reduced database is shown in Fig. 10. XGBoost had the highest prediction accuracy: RSME = 1.604, CoV = 0.114, Avg = 1.012, and IAE = 0.056. The reasons are that (i) XGBoost not only uses the first-order derivative but also uses the second-order derivative, which results in more accurate and customized loss function; (ii) the parallel optimization of XGBoost is based on feature granularity, and the default value of the branch can be specified for missing values or specified values, which greatly improves the efficiency; (iii) it supports column sampling, which mitigates overfitting and increases efficiency; and (iv) the parameter adjustment is flexible, and the model was adjusted according to the characteristics of the FRP-concrete interface test data. The performance of the SVM model is acceptable: RSME = 3.283, CoV = 0.209, Avg = 1.033, and IAE = 0.119. Although SVM is more often used in the case of a small scale database, it also shows strong performance with the reduced database in this study.

5.2. Comparison with existing equations

Fig. 11 shows that the ML models had higher accuracy than the existing equations in terms of RSME, CoV, Avg, and IAE. Regarding RSME, the top five performing models are all ML models, in the order of XGBoost, GBDT, ANN, RF, and DT. Model_14 [24] had the highest prediction accuracy among the 16 existing equations proposed in the literature. Regarding CoV, the top five performing models are XGBoost, GBDT, ANN, RF, and DT. Regarding Avg, the top five models are Model_15 [25], XGBoost, GBDT, ANN, and Model_12 [19]. Regarding IAE, the top five performing models are XGBoost, GBDT, ANN, RF, and SVM. The XGBoost model had the overall highest accuracy and was compared with Model_14 [24], which had the highest accuracy among the 16 existing equations. The RSME of the XGBoost model is 73% lower than that of Model_14. The CoV of the XGBoost model was 54% lower, and the IAE of the XGBoost model was 79% lower.

5.3. Importance of parameters

A parameter importance analysis was carried out on the parameters affecting the bond capacity of the FRP-concrete interface using the XGBoost model. After model training, the average gain for each parameter was generated, which was normalized to calculate the gain-based relative importance scale. Fig. 12 shows the analysis of the importance of the input parameters of b_c, L, f_c', K_f, and b_f on the output P_u. It can be seen that b_f has the highest weight (45.2%). K_f has the second largest (36.2%) contribution to P_u. It is interesting that L has the least contribution to P_u, which according to the authors' analysis, is due to the fact that most specimens included in the database have an L larger than the effective bond length.

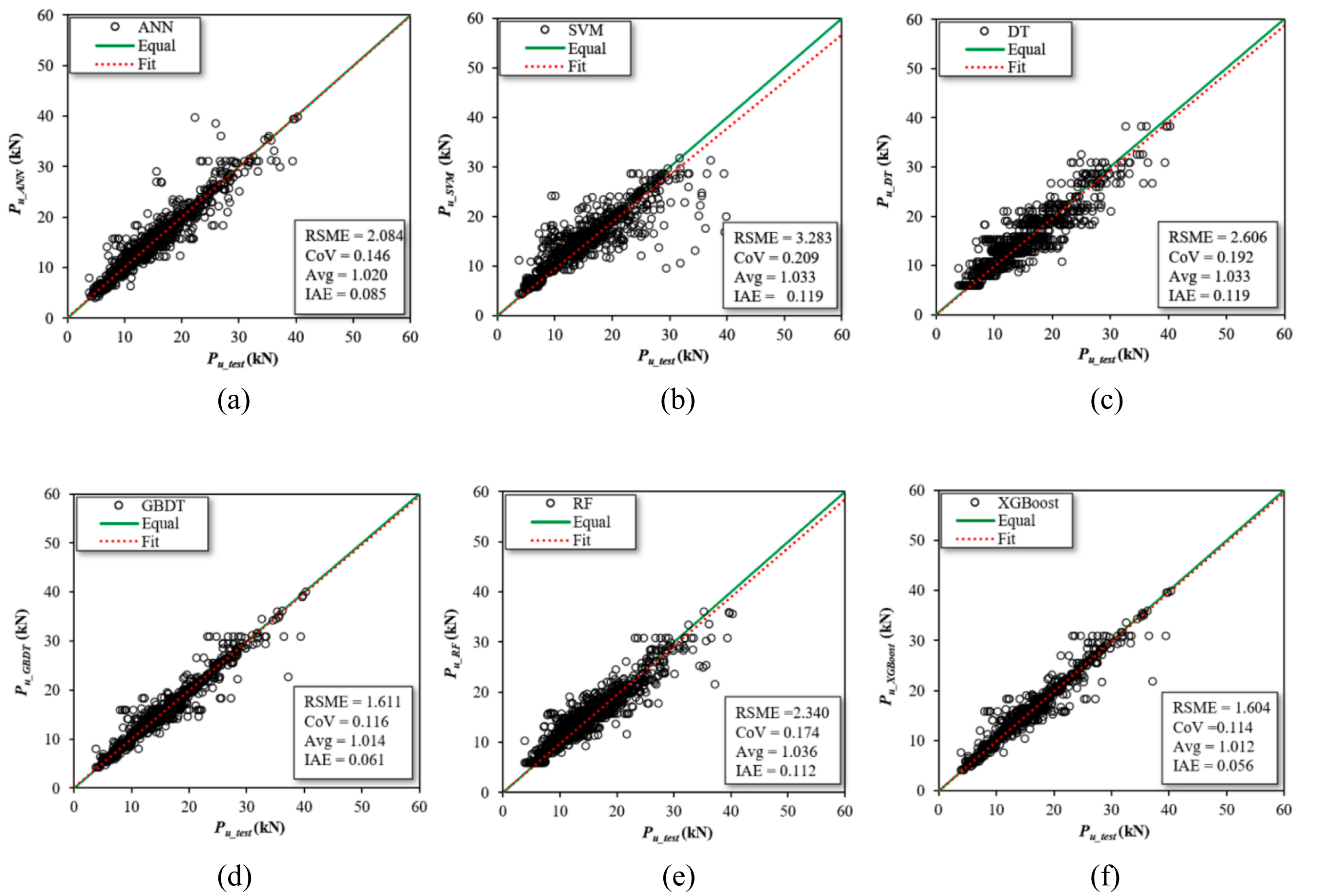

Fig. 10. Prediction performance of: (a) ANN, (b) SVM, (c) DT, (d) GBDT, (e) RF, and (f) XGBoost.

6. Formulation of equation

The above research showed that the ML models were able to predict the FRP-concrete interfacial bond strength. The models were used to formulate an explicit equation to facilitate the design of FRP-concrete structures. The form of the explicit equation was chosen to be similar to that of Eq. (9), the parameters of which have physical meaning. Eq. (9) includes a width coefficient to characterize the influence of the concrete shared force on the output P_u, so the term $\frac{b_f}{b_c}$ was used in the explicit equation. To avoid $\frac{b_f}{b_c}$ and b_f affecting the independence of variables, b_f was shifted to the left side of the equation.

$$\frac{P_u}{b_f} = F_1(K_f)F_2(f_c')F_3(L)F_4\left(\frac{b_f}{b_c}\right) \tag{10}$$

where $F_1(K_f) = K_f^{\alpha_1}$, $G_F = F_2(f_c') = \alpha_2(f_c')^{\beta_2}$, $\beta_L = F_3(L) = \alpha_3\tanh(\beta_3 L)$, and$\kappa_w = F_4\frac{b_f}{b_c} = 1 + \alpha_4\left[1 - \frac{b_f}{b_c}^{\beta_4}\right]$, and α_1, α_2, α_3, α_4, β_2, β_3, and β_4 are parameters to be determined based on the prediction data of the trained ML models.

Given the standard parameters, the trained ML model was used to predict the FRP-concrete interfacial bond strength for different combinations of input parameters. According to the reduced database, the standard value for each of the input parameters of Eq. (10) was mainly based on the median number of each parameter in the database, which is shown in Table 3.

The prediction curve displayed by tree models such as XGBoost algrithm is a discontinuous curve, which is attributed to the fact that the database has discrete points of each parameter. Thus, XGBoost is not suitable for curve formula fitting, and the well-trained ANN model was employed using the following steps:

First, the standard parameters were set as $f_c' = 45$ MPa, $L = 175$ mm, and $\frac{b_f}{b_c} = 0.4$ into the trained ANN model, and K_f was varied evenly over the database. The comparison between $\frac{P_u}{b_f}$ and K_f is shown in Fig. 13. The curve fitting of the predicted results of the $\frac{P_u}{b_f} - K_f$ relationship yields to:

$$G_1(K_f) = \frac{P_u}{b_f} = F_2(45)F_3(175)F_4(0.4)F_1(K_f) = 0.062K_f^{0.432} \tag{11}$$

The fitting has a R^2 of 0.994. It should be noted that when K_f is in the range of 150–200 GPa · mm and 300–375 GPa · mm, the fitting curve has notable discrepancy from the raw curve. Eq. (11) gives:

$$F_1(K_f) = \frac{G_1(K_f)}{F_2(45)F_3(175)F_4(0.4)} \tag{12}$$

Using the same procedure above, the relationship between $\frac{P_u}{b_f}$ and f_c' was fitted as:

$$G_2(f_c') = \frac{P_u}{b_f} = F_1(75)F_3(175)F_4(0.4)F_2(f_c') = 0.255f_c'^{0.119} \tag{13}$$

The comparison between Eq. (13) and the prediction data is shown in Fig. 14. The fitting has a R^2 of 0.984. Eq. (13) gives:

$$F_2(f_c') = \frac{G_2(f_c')}{F_1(75)F_3(175)F_4(0.4)} \tag{14}$$

Using the same procedure, the relationship between $\frac{P_u}{b_f}$ and L was fitted as:

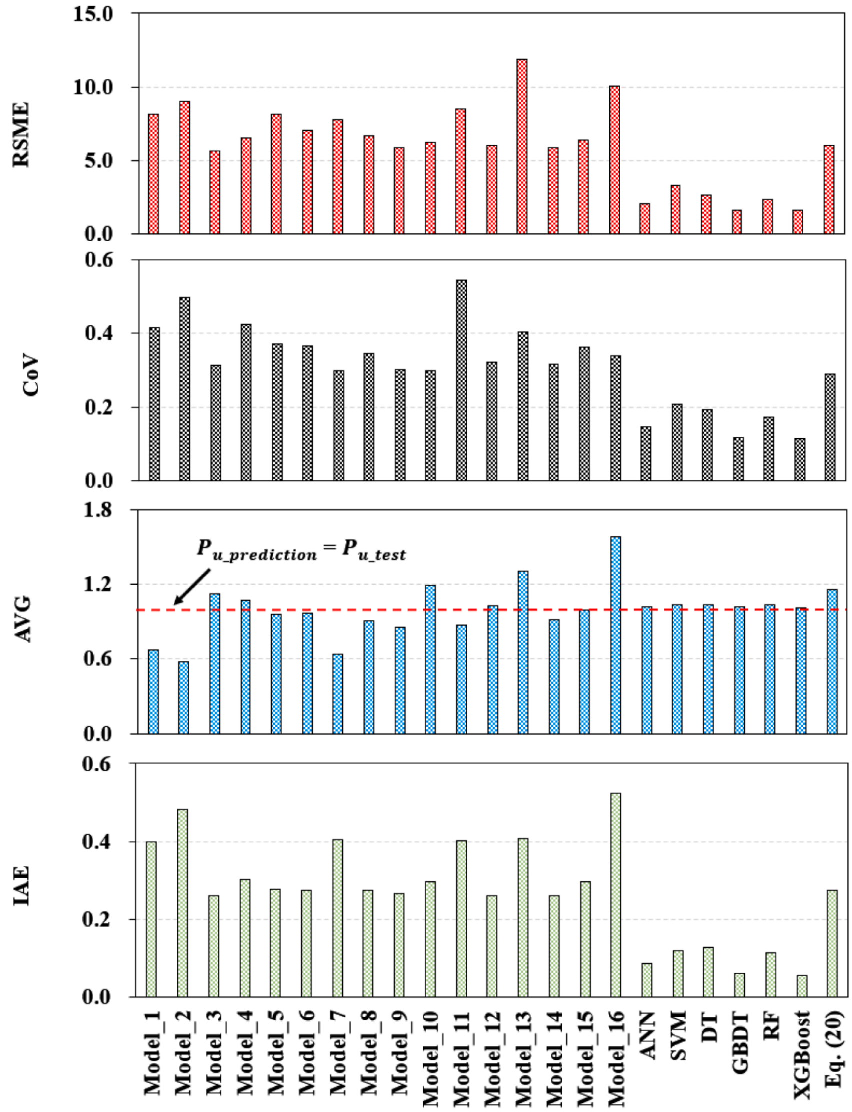

Fig. 11. Accuracy comparison of the existing equations and ML models.

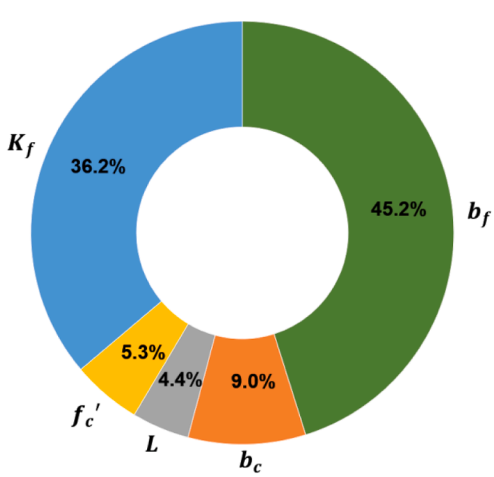

Fig. 12. Importance analysis of input parameters.

Table 3
Limit values and standard values for the input parameters.

Input variable	Minimum value	Maximum value	Standard value
f_c'(MPa)	5.4	75.5	45.0
L(mm)	20	700	175
b_f/b_c	0.05	1.00	0.40
K_f(GPa · mm)	10	360	75

$$G_3(L) = \frac{P_u}{b_f} = F_1(75)F_2(45)F_4(0.4)F_3(L) = 0.402\tanh(0.029L) \tag{15}$$

The comparison between Eq. (15) and prediction data is shown in Fig. 15. The R^2 is 0.886, which is relatively low. It can be found that for relatively short bonded length ($\leq$ 100 mm), the fitting curve has notable discrepancy from the raw curve.

$$F_3(L) = \frac{G_3(L)}{F_1(75)F_2(45)F_4(0.4)} \tag{16}$$

Using the same procedure, the relationship between $\frac{P_u}{b_f}$ and $\frac{b_f}{b_c}$ was fitted as:

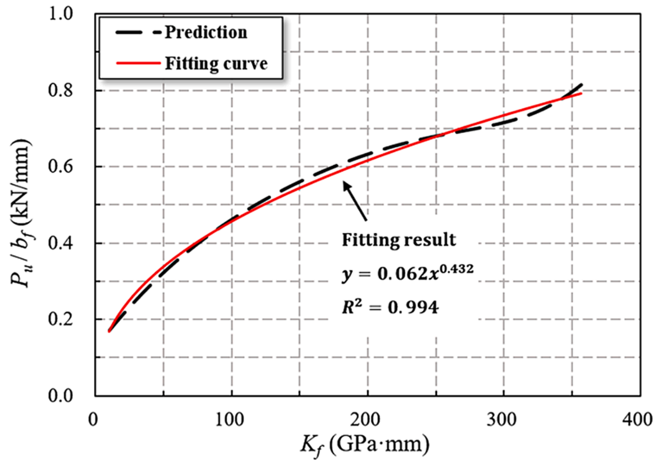

Fig. 13. Effect of stiffness of K_f on $\frac{P_u}{b_f}$.

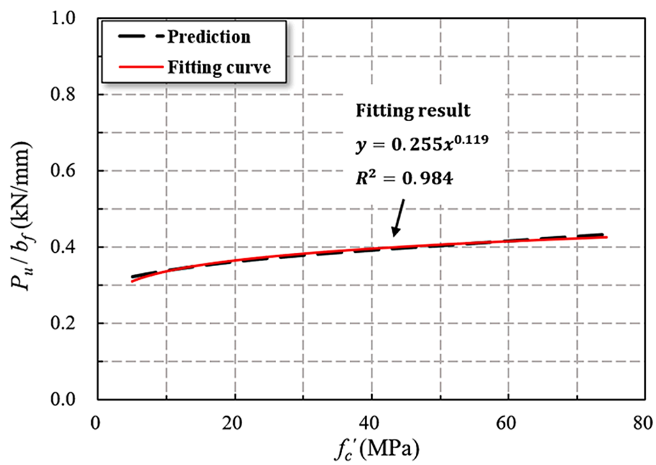

Fig. 14. Effect of f_c' on $\frac{P_u}{b_f}$.

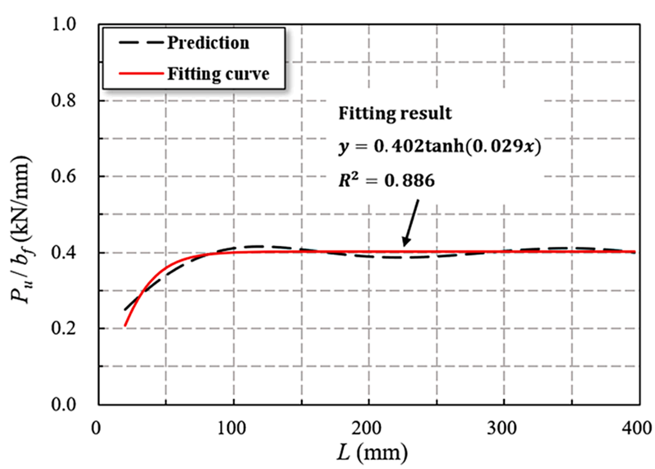

Fig. 15. Effect of L on $\frac{P_u}{b_f}$.

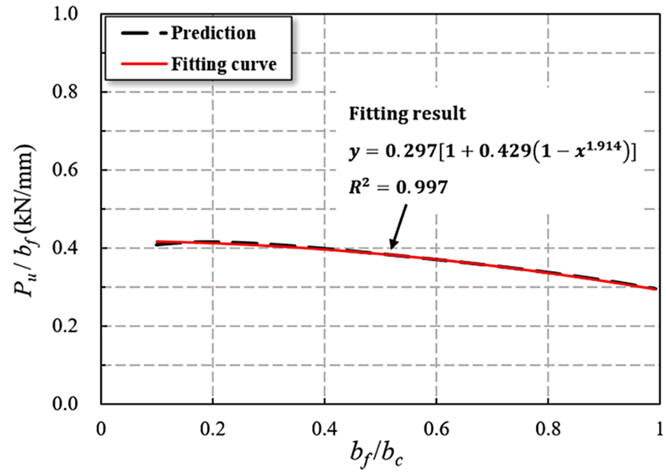

Fig. 16. Effect of $\frac{b_f}{b_c}$ on $\frac{P_u}{b_f}$.

$$G_4\left(\frac{b_f}{b_c}\right) = \frac{P_u}{b_f} = F_1(75)F_2(45)F_3(175)F_4\left(\frac{b_f}{b_c}\right) = 0.297\left[1 + 0.429\left(1 - \left(\frac{b_f}{b_c}\right)^{1.914}\right)\right] \tag{17}$$

The comparison between the Eq. (17) and the prediction data is shown in Fig. 16. The fitting has a R^2 of 0.997. Eq. (17) gives:

$$F_4\left(\frac{b_f}{b_c}\right) = \frac{G_4\left(\frac{b_f}{b_c}\right)}{F_1(75)F_2(45)F_3(175)} \tag{18}$$

Plugging Eqs. (12), (14), (16), (18) into Eq. (10) gives:

$$F_1(K_f)F_2(f_c')F_3(L)F_4\left(\frac{b_f}{b_c}\right) = \frac{G_1(K_f)G_2(f_c')G_3(L)G_4\left(\frac{b_f}{b_c}\right)}{F_1(75)^3F_2(45)^3F_3(175)^3F_4(0.4)^3} \tag{19}$$

Considering the standard value of the parameters (see Table 3), L= 175 mm, $\frac{b_f}{b_c} = 0.4$, f_c'= 45 MPa, and$K_f = 75$ GPa · mm, the result is $\frac{1}{F_1(75)^3F_2(45)^3F_3(175)^3F_4(0.4)^3} = 15.485$. Finally, Eq. (20) was obtained:

$$P_u = 15.485G_1(K_f)G_2(f_c')G_3(L)G_4\left(\frac{b_f}{b_c}\right)b_f \tag{20}$$

Fig. 17 shows the performance of Eq. (20) to predict the test results of

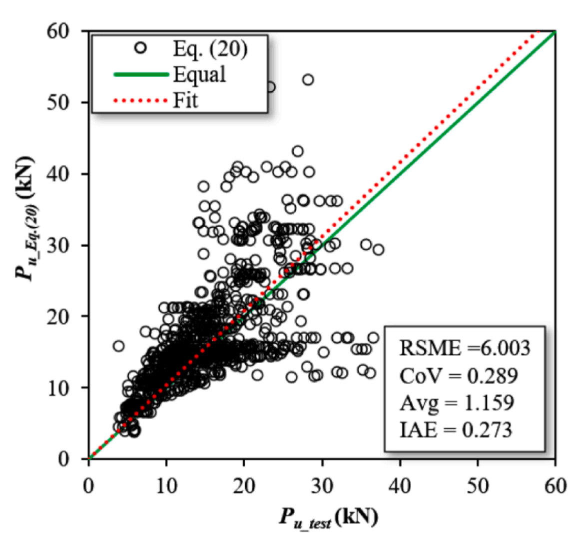

Fig. 17. Prediction performance of Eq. (20).

the specimens included in the reduced database. The results are also included in Fig. 11, where RSME of Eq. (20) is 6.003, CoV is 0.289, Avg is 1.159, and IAE is 0.273. Although the accuracy of the prediction results is lower than that of the original ANN model, nevertheless, Eq. (20) has high accuracy compared with the 16 models from the literature. It can be seen that Eq. (20) can give more accurate results, avoid the “black box” of ANN, and enable users to see the internal work process data interpretation.

7. Conclusions

This study proposed a practical equation to predict the bonding strength of the FRP-concrete interface based on the proposed well-trained ML model and traditional analytical methods considering the physical meaning of the parameters. A large database was established by collecting experimental test data from different studies reported in the literature. The database contains the test results from 1375 FRP-concrete direct shear specimens. The performance of six ML methods was compared to optimize the predictive model. A new equation, with parameters associated with physical meanings, was generated to predict the interfacial bond strength by considering the influencing parameters identified from ML and combining the form of existing prediction formulas. Based on the results and analysis, the following conclusions are drawn:

(1) Analyzing the correlation of six variables in the database revealed that there is a high correlation between f_t and $f_c^{'}$, with a Pearson correlation coefficient as high as 0.931. Therefore, five influencing factors (L, $f_c^{'}$, b_c, b_f, K_f) affecting the FRP-concrete interfacial bond strength were analyzed, in which b_f and K_f were determined to have the greatest influence on the interfacial bond strength.

(2) Sixteen traditional physical-meaning based empirical equations proposed by previous researchers to predict the FRP-concrete interfacial bond strength were evaluated using the database. These equations showed limited accuracy and variation. Among them, the model by Lu et al. (Model_14) [24] had the best performance.

(3) Six ML algorithms were used to train and predict the experimental data collected in this paper. The ML models substaintially increased the prediction accuracy compared with the 16 traditional empirical equations, and they especially reduced the variation. Among them, XGBoost had the best performance. Compared with the model by Lu et al. (Model 14) [24], the XGBoost model reduced the RSME by 72.7%, reduced the CoV by 53.6%, and reduced the IAE by 78.5%.

(4) In order to solve the black-box problem of ML models and provide guidance for practical design, an explicit and practical equation, Eq. (20) in this paper, was derived based on the ANN model to predict the FRP-concrete interfacial bond strength, which showed high prediction accuracy. The proposed equation combines the accurate ANN model and has physical meaning as it is presented in a form that is similar to existing equations in the literature.

The current research provides a method for the interpretability study of ML models. The accuracy of the proposed equation can be further improved if the fitting of results in Figs. 13 and 15 can be improved. Results can also be improved as the interfacial debonding mechanism is further revealed and more test data are incorporated.

CRediT authorship contribution statement

Feng Zhang: Writing – original draft. **Chenxin Wang:** . **Jun Liu:** Conceptualization, Methodology. **Xingxing Zou:** Conceptualization, Data curation, Funding acquisition, Methodology, Supervision, Writing – review & editing. **Lesley H. Sneed:** Writing – review & editing, Methodology. **Yi Bao:** Conceptualization, Methodology, Writing – review & editing. **Libin Wang:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgement

This study was funded by National Natural Science Foundation of China (No. 52208256), Basic Research Program from the National Science Foundation of Jiangsu Province, China (No. BK20210620) and Natural Science Research of Jiangsu Higher Education Institutions of China (No. 21KJB560007). Many thanks also go to reviewers for their constructive suggestions.

Availability of data and materials.

The database is available upon requested from the corresponding author of this study.

Appendix A. Existing prediction equations for interfacial bond strength

Ref.	Formula						Model
	P_u(kN)	L_e(mm)	τ_a(MPa)	κ_w	G_F(MPa · mm)	β_L	
[99]	$b_f\sqrt{G_F K_f}$				$0.204f_t$		1
[16]	$b_f\tau_a L$		$6.13-lnL$				2
[27]	$0.64\kappa_w b_f\beta_L\sqrt{f_t K_f}$	$\sqrt{\frac{K_f}{2f_t}}$		$\sqrt{1.125\frac{2-b_f/b_c}{1+b_f/400}}$		$\begin{cases}1 & if\ L\geq L_e\\ \frac{L}{L_e}\left(2-\frac{L}{L_e}\right) & if\ L<L_e\end{cases}$	3
[17]	$b_f\tau_a L$		$5.88L^{-0.669}$				4
[18]	$b_f\tau_a\beta_L L$	$e^{6.13-0.58ln(K_f)}$	$110.2\times 10^{-6}E_f t_f$			$\begin{cases}1 & if\ L\geq L_e\\ \frac{L_e}{L} & if\ L<L_e\end{cases}$	5
[20]	$b_f\tau_a\beta_L L$	$e^{6.13-0.58ln(K_f)}$					6

(continued on next page)

(*continued*)

Ref.	Formula						Model
	$P_u(\mathrm{kN})$	$L_e(\mathrm{mm})$	$\tau_a(\mathrm{MPa})$	κ_w	$G_F(\mathrm{MPa}\cdot\mathrm{mm})$	β_L	
			$110.2\times 10^{-6}K_f\left(\frac{f_{cu}}{42}\right)^{2/3}$			$\begin{cases}1 & if\ L\geq L_e\\ \frac{L_e}{L} & if\ L<L_e\end{cases}$	
[100]	$0.78b_f\beta_L\sqrt{\kappa_w G_F K_f}$	$\sqrt{\frac{K_f}{4f_t}}$		$\sqrt{1.125\frac{2-b_f/b_c}{1+b_f/400}}$	$0.204f_t$	$\begin{cases}1 & if\ L\geq L_e\\ \frac{L}{L_e}\left(2-\frac{L}{L_e}\right) & if\ L<L_e\end{cases}$	7
[21]	$\left(0.5+0.008\sqrt{\frac{K_f}{f_t}}\right)\tau_a b_f L_e$	100	$0.5f_t$				8
[1]	$0.427b_f\beta_L L\kappa_w\sqrt{f_{cu}}$	$\sqrt{\frac{K_f}{\sqrt{f_c'}}}$		$\sqrt{\frac{2-b_f/b_c}{1+b_f/b_c}}$		$\begin{cases}\frac{L_e}{L} & if\ L\geq L_e\\ \frac{L_e}{L}\sin\frac{\pi L}{2L_e} & if\ L<L_e\end{cases}$	9
[22]	$b_f\frac{\sqrt{E_f t_f\tau_a}}{3}$	$\sqrt{\frac{K_f}{\sqrt{4\tau_a}}}$	$1.8k_w f_t$	$\sqrt{1.5\frac{2-b_f/b_c}{1+b_f/100}}$		$\begin{cases}1 & if\ L\geq L_e\\ sin\frac{\pi L}{2L_e} & if\ L<L_e\end{cases}$	10
[23]	$f_c'^{\,0.2}b_f\beta_L L$	$0.7\sqrt{\frac{K_f}{f_c'^{\,0.2}}}$				$\begin{cases}1.1\frac{L}{L_e} & if\ L>L_e\\ \left[0.7cos\left(\frac{L_b}{L_e}\right)^{\pi}+1.8\right] & if\ L<L_e\end{cases}$	11
[19]	$\tau_a b_f\beta_L L$	$0.125(K_f)^{0.57}$	$0.93f_{cu}^{0.44}$			$\begin{cases}1 & if\ L\geq L_e\\ \frac{L_e}{L} & if\ L<L_e\end{cases}$	12
[101]	$\kappa_w b_f\sqrt{2K_f G_F}$	$0.74\sqrt{\frac{K_f}{f_{cu}^{0.236}}}$		$\begin{cases}1, & b_f<100\\ 1+\frac{7.4}{b_f}, & b_f\geq 100\end{cases}$	$0.514f_{co}^{\,0.236}$		13
[24]	$\kappa_w b_f\beta_L\sqrt{2K_f\kappa_w G_F}$	$L_e=a_o+\frac{1}{2\lambda_1}ln\frac{\lambda_1+\lambda_2\tan(\lambda_2 a_o)}{\lambda_1-\lambda_2\tan(\lambda_2 a_o)}a_0=\frac{1}{\lambda_2}\sin^{-1}\left(0.99\sqrt{\frac{s_f-s_0}{s_f}}\right)s_f=\frac{2G_F}{\tau_{max}},\ s_0=0.0195k_w f_t$ $\lambda_1=\sqrt{\frac{\tau_{max}}{s_0 E_f t_f}},\ \lambda_2=\sqrt{\frac{\tau_{max}}{(s_f-s_0)E_f t_f}}$	$1.5f_t$	$\sqrt{\frac{2.25-b_f/b_c}{1.25+b_f/b_c}}$	$0.308\sqrt{f_t}$	$\begin{cases}1, & L\geq L_e\\ \frac{L}{L_e}\left(2-\frac{L}{L_e}\right), & L<L_e\end{cases}$	14
[25]	$0.585b_f f_{cu}^{0.1}k_w(K_f)^{0.54}\beta_L$	$\frac{0.395(K_f)^{0.54}}{f_{cu}^{0.09}}$		$\sqrt{\frac{2.25-b_f/b_c}{1.25+b_f/b_c}}$		$\begin{cases}1, & L\geq L_e\\ \left(\frac{L}{L_e}\right)^{1.2}, & L<L_e\end{cases}$	15
[26]	$\beta_L b_f\kappa_w\sqrt{2K_f G_F}$	$1.6841\sqrt{\frac{K_f}{f_c^{2/3}}}$		$\sqrt{\frac{2.9-b_f/b_c}{0.6+b_f/b_c}}$	$0.0498\sqrt{f_c}$	$\begin{cases}1, & L\geq L_e\\ \frac{L}{L_e}\left(2-\frac{L}{L_e}\right), & L<L_e\end{cases}$	16

References

[1] Chen JF, Teng JG. Anchorage strength models for FRP and steel plates bonded to concrete. J Struct Eng 2001;127(7):784–1779.

[2] Ceroni F, Pecce M. Cracking behaviour of RC beams externally strengthened with emerging materials. Constr Build Mater 2007;21(4):736–45.

[3] Bakis CE, Bank LC, Brown V, Cosenza E, Davalos JF, Lesko JJ, et al. Fiber-reinforced polymer composites for construction—State-of-the-art review. J Compos Constr 2002;6(2):73–87.

[4] Buyukozturk O, Hearing B. Failure behavior of precracked concrete beams retrofitted with FRP. J Compos Constr 1998;2(3):138–44.

[5] Sneed LH, Verre S, Carloni C, Ombres L. Flexural behavior of RC beams strengthened with steel-FRCM composite. Eng Struct 2016;127:686–99.

[6] Yun Y, Wu YF, Tang WC. Performance of FRP bonding systems under fatigue loading. Eng Struct 2008;30(11):3129–40.

[7] Hollaway LC. A review of the present and future utilisation of FRP composites in the civil infrastructure with reference to their important in-service properties. Constr Build Mater 2010;24(12):2419–45.

[8] Smith ST, Teng JG. FRP-strengthened RC beams. I: review of debonding strength models. Eng Struct 2002;24(4):385–95.

[9] Ross CA, Jerome DM, Tedesco JW, Hughes ML. Strengthening of reinforced concrete beams with externally bonded composite laminates. Structural Journal 1999;96(2):212–20.

[10] Wu YF, Zhou ZQ, Yang QD, Chen WQ. On shear bond strength of FRP-concrete structures. Eng Struct 2010;32(3):897–905.

[11] Chaallal O, Nollet MJ, Perraton D. Strengthening of reinforced concrete beams with externally bonded fiber. Can J Civ Eng 1998;25(4):692–704.

[12] Carloni C, Subramaniam KV, Savoia M, Mazzotti C. Experimental determination of FRP–concrete cohesive interface properties under fatigue loading. Compos Struct 2012;94(4):1288–96.

[13] Zhou YW, Zheng SB, Huang ZY, Sui LL, Chen Y. Explicit neural network model for predicting FRP-concrete interfacial bond strength based on a large database. Compos Struct 2020;240:111998.

[14] Liu K, Wu YF. Analytical identification of bond–slip relationship of EB-FRP joints. Compos B Eng 2012;43(4):1955–63.

[15] Pan J, Wu YF. Analytical modeling of bond behavior between FRP plate and concrete. Compos B Eng 2014;61:17–25.

[16] Tanaka T. Shear resisting mechanism of reinforced concrete beams with CFS as shear reinforcement. Hokkaido University, Hokkaido, Japan; (1996).

[17] Yoshizawa H. Analysis of debonding fracture properties of CFS strengthened RC member subject to tension. Proceedings of the 3rd International Symposium on Non-Metallic (FRP) Reinforcement for Concrete Structures, Sapporo, Japan, 287–94 (1997).

[18] Maeda T, Asano Y, Sato Y, Ueda T, Kakuta Y. A study on bond mechanism of carbon fiber sheet. Proceedings of the 3rd International Symposium on Non-Metallic (FRP) Reinforcement for Concrete Structures, Sapporo, Japan, 1, 279–85 (1997).

[19] Japan Concrete Institute (2003). Technical report of technical committee on retrofit technology. Proceedings of International Symposium on the Latest Achievement of Technology and Research on Retrofitting Concrete Structures, Kyoto, Japan.
[20] Khalifa A, Gold WJ, Nanni A, Mi AA. Contribution of externally bonded FRP to shear capacity of RC flexural members. J Compos Constr 1998;2(4):195–202.
[21] Yang YX, Yue QR, Hu YC. Experimental study on bond performance between carbon fiber sheets and concrete. J Build Struct 2001;22(3):36–42.
[22] Monti G, Renzelli M, Luciani P. FRP adhesion in uncracked and cracked concrete zones. Fibre-Reinforc Polym Reinforce Concr Struct 2003;2:183–92.
[23] Kanakubo T, Furuta T, Fukuyama H. Bond strength between fiber-reinforced polymer laminates and concrete. In: Fibre-Reinforced Polymer Reinforcement for Concrete Structures, 2; 2003. p. 133–42.
[24] Lu XZ. Studies on FRP-concrete interface. Tsinghua University, Beijing, China; (2004). [in Chinese].
[25] Wu Z, Islam SM, Said H. A three-parameter bond strength model for FRP-concrete interface. J Reinf Plast Compos 2009;28(19):2309–23.
[26] Zhou YW. Analytical and experimental study on the strength and ductility of FRP-reinforced high strength concrete beam. Dalian University of Technology, Dalian, China; (2009). [in Chinese].
[27] Neubauer U, Rostasy FS. Design aspects of concrete structures strengthened with externally bonded CFRP-plates. Proceedings of the Seventh International Conference on Structural Faults and Repair, Edinburgh, Scotland, 2, 109–18 (1997).
[28] Subramaniam KV, Carloni C, Nobile L. Width effect in the interface fracture during shear debonding of FRP sheets from concrete. Eng Fract Mech 2007;74(4): 578–94.
[29] Zhang W, Tang Z, Yang Y, Wei J. Assessment of FRP–concrete interfacial debonding with coupled mixed-mode cohesive zone model. J Compos Constr 2021;25(2):04021002.
[30] Zhang W, Huang Y. Three-dimensional numerical investigation of mixed-mode debonding of FRP-concrete interface using a cohesive zone model. Constr Build Mater 2022;350:128818.
[31] Chen SZ, Zhang SY, Han WS, Wu G. Ensemble learning based approach for FRP-concrete bond strength prediction. Constr Build Mater 2021;302:124230.
[32] Su M, Zhong Q, Peng H, Li S. Selected machine learning approaches for predicting the interfacial bond strength between FRPs and concrete. Constr Build Mater 2021;270:121456.
[33] Jahangir H, Eidgahee DR. A new and robust hybrid artificial bee colony algorithm–ANN model for FRP-concrete bond strength evaluation. Compos Struct 2021;257:113160.
[34] Vu DT, Hoang ND. Punching shear capacity estimation of FRP-reinforced concrete slabs using a hybrid machine learning approach. Struct Infrastruct Eng 2016;12 (9):1153–61.
[35] Mahjoubi S, Meng W, Bao Y. Logic-guided neural network for predicting steel-concrete interfacial behaviors. Expert Syst Appl 2022;198:116820.
[36] Mahjoubi S, Meng W, Bao Y. Auto-tune learning framework for prediction of flowability, mechanical properties, and porosity of ultra-high-performance concrete (UHPC). Appl Soft Comput 2022;115:108182.
[37] Mahjoubi S, Barhemat R, Guo P, Meng W, Bao Y. Prediction and multi-objective optimization of mechanical, economical, and environmental properties for strain-hardening cementitious composites (SHCC) based on automated machine learning and metaheuristic algorithms. J Cleaner Prod 2021;329:129665.
[38] Guo P, Meng W, Xu M, Li V, Bao Y. Predicting mechanical properties of high-performance fiber-reinforced cementitious composites by integrating micromechanics and machine learning. Materials 2021;14(12):3143.
[39] Taljsten B. (1994). Plate bonding: strengthening of existing concrete structures with epoxy bonded plates of steel or fiber reinforced plastics. Lulea University of Technology, Scandinavia, Sweden.
[40] Chajes MJ, Finch WW, Thomson TA. Bond and force transfer of composite-material plates bonded to concrete. Struct J 1996;93(2):209–17.
[41] Yao J. Debonding in FRP-Strengthened RC Structures. The Hong Kong Polytechnic University, Hong Kong, China; (2004).
[42] Takeo K, Matsushita H, Makizumi T, Nagashima G. Bond characteristics of CFRP sheets in the CFRP bonding technique. Proceedings of Japan Concrete Institute 1997;19(2):1599–604.
[43] Zhao HD, Zhang Y, Zhao M. Experimental study on axial compression of reinforced concrete cylinder reinforced with carbon fiber reinforced polymer. Build Struct 2000;307. 5.[in Chinese].
[44] Ueda T, Sato Y, Asano Y. Experimental study on bond strength of continuous carbon fiber sheet. Spec Publ 1999;188:407–16.
[45] Nakaba K, Toshiyuki K, Tomoki F, Hiroyuki Y. Bond behavior between fiber-reinforced polymer laminates and concrete. Struct J 2001;98(3):359–67.
[46] Wu ZS, Yuan H, Hiroyuki Y, Toshiyuki K. Experimental/analytical study on interfacial fracture energy and fracture propagation along FRP-concrete interface. Spec Publ 2001;201:133–52.
[47] Ren HT. Study on basic theories and long time behavior of concrete structures strengthened by fiber reinforced polymer. Dalian University of Technology, Dalian, China. [in Chinese] (2003).
[48] Antonio B, Marco DL, Emidio N. FRP-to-concrete interface debonding: experimental calibration of a capacity model. Compos B Eng 2011;42:1539–53.
[49] Savoia M, Ferracuti B. Strengthening of RC structure by FRP: experimental analyses and numerical modelling. University of Bologna, Bologna, Italy (2006).
[50] Savoia M, Bilotta A, Ceroni F, Di L M, Fava G, Ferracuti B, Mazzotti C, Nigro E, Olivito R, Pecce M, Poggi C. Experimental round robin test on FRP-concrete bonding. Proceedings of the 9th International Symposium on Fiber Reinforced Polymer Reinforcement for Concrete Structures, Sydney, Australia, 13-15 (2009).
[51] Huy BP, Riadh A. Modelling of CFRP-concrete shear-lap tests. Constr Build Mater 2007;21(4):727–35.
[52] Liu J. (2012). Effect of concrete strength on interfacial bond behavior of CFRP-concrete. Shenzhen University, Shenzhen, China. [in Chinese].
[53] Toutanji H, Saxena P, Zhao L, Ooi T. Prediction of interfacial bond failure of FRP concrete surface. J Compos Constr 2007;11(4):427–36.
[54] Bilotta A, Ceroni F, Di LM, Nigro E, Pecce M, Manfredi G. Bond efficiency of EBR and NSM FRP systems for strengthening of concrete members. J Compos Constr 2011;15(5):757–72.
[55] Faella C, Nigro E, Martinelli E, Sabatino M, Salerno N, Mantegazza G. Aderenza tra calcestruzzo e Lamine di FRP utilizzate come placcaggio di elementi inflessi. Parte I: Risultati sperimentali. In XIV Congresso CTE, Mantova, Italy, 7-8. [In Italian] (2002).
[56] Woo SK, Lee Y. Experimental study on interfacial behavior of CFRP-bonded concrete. KSCE J Civ Eng 2010;14(3):385–93.
[57] Wu YF, Jiang C. Quantification of bond-slip relationship for externally bonded FRP to-concrete joints. J Compos Constr 2013;17(5):673–86.
[58] Dai JG, Sato Y, Ueda T. Improving the load transfer and effective bond length for FRP composites bonded to concrete. Proceed Japan Concr Inst 2002;24(1): 1423–8.
[59] Tan Z. Experimental study on the performance of concrete beam strengthened with GFRP. Tsinghua University, Beijing, China. [in Chinese] (2002).
[60] Sharma SK, Mohamed AMS, Goldar D, Sikdar PK. Plate-concrete interfacial bond strength of FRP and metallic plated concrete specimens. Compos B Eng 2006;37: 54–63.
[61] Fen ZL, Gu XL, Zhang WP, Liu LM. Experimental study on bond behavior between carbon fiber reinforced polymer and concrete. Struct Eng 2008;24(4):154–63 [in Chinese].
[62] Adhikary B, Mutsuyoshi H. Study on the bond between concrete and externally bonded CFRP sheet. Proceedings of the 5th International Symposium On Fiber Reinforced Concrete Structures, Cambridge, UK, 371–378 (2001).
[63] Fu Q, X U, Jian G G, Yu C. Bond strength between CFRP sheets and concrete. In FRP Composites in Civil Engineering: Proceedings of the International Conference on FRP Composites in Civil Engineering, Beijing, China, 1, 357 (2001).
[64] Ueno S, Toutanji H, Vuddandam R. Introduction of a stress state criterion to predict bond strength between FRP and concrete substrate. J Compos Constr 2015;19(1):04014024.
[65] Zhang HW, Smith ST. Fiber-reinforced polymer (FRP)-to-concrete joints anchored with FRP anchors: test and experimental trends. Can J Civ Eng 2013;40(11): 1103–16.
[66] Ko H, Matthys S, Palmieri A, Sato Y. Development of a simplified bond stress-slip model for bonded FRP-concrete interfaces. Constr Build Mater 2014;68(4): 142–57.
[67] Carlo P, Davide T, Claudio M. Experimental study on bond behavior between concrete and FRP reinforcement. J Compos Constr 2008;12(2):180–9.
[68] Kamiharako A, Shimomura T, Maruyama K. The Influence of the substrate on the bond behavior of continuous fiber sheet. Proceedings of Japan Concrete Institute 2003;25(2):1735–40 [in Japanese].
[69] Kanakubo T, Nakaba K, Furuta T, Yoshizawa H. A proposal of local stress–slippage relationship between fiber sheet and concrete. Concr Res Technol 2001;12(1):33–43.
[70] Czaderski C, Olia S. EN-CORE round robin testing program-contribution of EMPA. In Proceedings of 6th International Conference on FRP Composites in Civil Engineering, Rome, Italy (2012).
[71] Hosseini A, Mostofinejad D. Effective bond length of FRP-to-concrete adhesively bonded joints: experimental evaluation of existing models. Int J Adhes Adhes 2014;48:150–8.
[72] Zou X, Sneed LH. Bond behavior between steel fiber reinforced polymer (SRP) and concrete. Int J Concr Struct Mater 2020;14(1):1–17.
[73] Santandrea M. Bond behavior between fiber reinforced composites and quasi-brittle material interfaces.University of Bologna, Bologna, Italty (2018).
[74] Ascione F, Lamberti M, Napoli A, Razaqpur G, Realfonzo R. An experimental investigation on the bond behavior of steel reinforced polymers on concrete substrate. Compos Struct 2017;181:58–72.
[75] Liu W, Liu C, Liu M, Xu F, Li Z. Investigation on interfacial properties and calculation models of bamboo scrimber-to-concrete bonding joint. Constr Build Mater 2021;313:125530.
[76] Zhang F, Wan B, Xu Q, Wang M, Wang Z, Leng Y. Experimental study of bond behavior of laminated bamboo plate to concrete interfaces. Mater Struct 2021;54 (1):1–14.
[77] Ai BW. Experimental study on interface characteristics between CFRP and concrete. East China Jiaotong University, Shanghai, China. [in Chinese] (2016).
[78] Biolzi L, Ghittoni C, Fedele R, Rosati G. Experimental and theoretical issues in FRP-concrete bonding. Constr Build Mater 2013;41:182–90.
[79] Shi JW, Cao WH, Wu ZS. Effect of adhesive properties on the bond behaviour of externally bonded FRP-to-concrete joints. Compos B Eng 2019;177:107365.
[80] Barbieri G, Biolzi L, Bocciarelli M, Cattaneo S. Size and shape effect in the pull-out of FRP reinforcement from concrete. Compos Struct 2016;143:395–417.
[81] Serbescu A, Guadagnini M, Pilakoutas K. Standardised double-shear test for determining bond of FRP to concrete and corresponding model development. Compos B Eng 2013;55:277–97.
[82] Pan J, Leung CK. Effect of concrete composition on FRP/concrete bond capacity. J Compos Constr 2007;11(6):611–8.

[83] Cao SY, Chen JF, Pan JW, Sun N. ESPI measurement of bond-slip relationships of FRP-concrete interface. J Compos Constr 2007;11(2):149–60.
[84] Chen C, Li X, Zhao D, Huang Z, Sui L, Xing F, et al. Mechanism of surface preparation on FRP-concrete bond performance: a quantitative study. Compos B Eng 2019;163:193–206.
[85] Mostofinejad D, Arefian B. Generic assessment of effective bond length of FRP-concrete joint based on the initiation of debonding: Experimental and analytical investigation. Compos Struct 2021;277:114625.
[86] Mostofinejad D, Sanginabadi K, Eftekhar MR. Effects of coarse aggregate volume on CFRP-concrete bond strength and behavior. Constr Build Mater 2019;198:42–57.
[87] Carloni C, Verre S, Sneed LG, Ombres L. Loading rate effect on the debonding phenomenon in fiber reinforced cementitious matrix-concrete joints. Compos B Eng 2017;108:301–14.
[88] ASTM C39/C39M-17b. Standard test method for compressive strength of cylindrical concrete specimens. West Conshohocken, Pam: ASTM International (2017).
[89] Guo ZH. Principles of reinforced concrete. Butterworth-Heinemann (2014).
[90] EN 206: 2013+A1, Concrete, Specification, Performance, Production and Conformity, European Committee for Standardization, Brussels, 2016.
[91] GB 50010. Code for design of concrete structures. Beijing, China, Building Industry Press, 2002. [In Chinese].
[92] Bamforth, P., Chisholm, D., Gibbs, J., Harrison, T. (2008). Properties of concrete for use in Eurocode 2.
[93] Candelieri A. Clustering and support vector regression for water demand forecasting and anomaly detection. Water 2017;9(3):224.
[94] Liu FT, Ting KM, Zhou ZH. Isolation forest. In: In 2008 Eighth Ieee International Conference on Data Mining; 2008. p. 413–22.
[95] Hornik K, Stinchcombe M, White H. Multilayer feedforward networks are universal approximators. Neural Netw 1989;2(5):359–66.
[96] Zhou ZH. Machine learning. Springer Nature Singapore; 2021.
[97] Geyer R, Jambeck JR, Law KL. Production, use, and fate of all plastics ever made. Sci Adv 2017;3(7):e1700782.
[98] Chen T, Guestrin C. Xgboost: A scalable tree boosting system. In Proceedings of the 22nd ACM Sigkdd International Conference on Knowledge Discovery and Data Mining, San Francisco, USA, 785–794 (2016).
[99] Holzenkämpfer P. Ingenieurmodelle des verbunds geklebter bewehrung für betonbauteile. IBMB, Bonn, Germany (1994).
[100] Niedermeier R. Envelope line of tensile forces while using externally bonded reinforcement. Technical University of Munich, Munich, Germany (2000).
[101] Dai J, Ueda T, Sato Y. Development of the nonlinear bond stress-slip model of fiber reinforced plastics sheet-concrete interfaces with a simple method. J Compos Constr 2005;9(1):52–62.